
NISOLON TABLET 5 MG

DESCRIPTION:

A white, round shape tablet, 7 mm in diameter with marking 'd' and scored on another side.

COMPOSITION:

Each tablet contains Prednisolone 5 mg.

PHARMACODYNAMICS:

Prednisolone is a synthetic corticosteroid with glucocorticoid and anti-inflammatory effects. Prednisolone exceeds hydrocortisone in glucocorticoid and anti-inflammatory activity, being about three times more potent on a weight basis than the parent hormone, but is considerably less active than hydrocortisone in mineralocorticoid activity.

Prednisolone like hydrocortisone is a potent therapeutic agent influencing the biochemical behaviour of most tissues of the body.

The mechanism of action of corticosteroids is thought to be by control of protein synthesis. Corticosteroids react with receptor proteins in the cytoplasm of sensitive cells in many tissues to form a steroid- receptor complex.

Corticosteroids are palliative symptomatic treatment by virtue of their anti-inflammatory effects; they are never curative.

PHARMACOKINETICS:

Prednisolone is readily absorbed from the gastrointestinal tract.

Absorption: The plasma half-life after oral administration of prednisolone ranges from 3 to 4 hours. Oral bioavailability varies widely between subjects. Peak plasma concentrations of prednisolone are obtained 1 or 2 hours after oral administration and prednisolone has a usual plasma half-life of 2 to 4 hours. Its initial absorption, but not its overall bioavailability, is affected by food.

Distribution: Prednisolone is 90 to 95% bound to plasma proteins.

Metabolism: Prednisolone is conjugated in the liver and to some extent in the kidney.

Excretion: Prednisolone is excreted in the urine as free and conjugated metabolites, together with an appreciable proportion of unchanged prednisolone. Prednisolone crosses the placenta and small amounts are excreted in breast milk.

INDICATION:

Collagen diseases: (eg. rheumatoid arthritis, acute rheumatic fever, SLE).

Allergic diseases: (such as severe bronchial asthma, intractable allergic rhinitis, exfoliative dermatitis, serum sickness, and other acute drug reactions). Generalised dermatosis having an allergic component (eg. contact dermatitis, neurodermatitis, allergic eczema).

Acute ocular inflammatory disease involving structures in the posterior segment (such as uveitis, chorioretinitis in vital areas, sympathetic ophthalmia). Certain lymphatic neoplastic diseases (to induce temporary remission).

Various chronic or recurrent diseases of unknown aetiology such as ulcerative colitis and nephrosis.

Since Nisolon lacks significant mineralocorticoid activity in usual therapeutic doses, it is not adequate for acute adrenocortical insufficiency. Adrenocortical steroids, hydrocortisone and cortisone should be used. Whenever corticosteroid therapy is indicated

RECOMMENDED DOSAGE:

Usual adult and adolescent doses:

Oral, 5 to 60mg a day as a single dose or in divided doses. For acute exacerbations of multiple sclerosis Oral, 200 mg per day for one week, followed by 80mg every other day for one month

Usual adult prescribing limits:

Oral, up to 250mg a day

Usual pediatric dose:

Adrenocortical insufficiency – Oral, 140mcg (0.14mg) per kg of body weight or 4 mg per square meter of body surface a day in three divided doses.

Other indications – Oral, 500 mcg (0.5mg) to 2 mg per kg of body weight or 15 to 60 mg per square meter of body surface a day in three or four divided doses

ROUTE OF ADMINISTRATION: For oral use**CONTRAINDICATIONS:**

Systemic fungal infections and known hypersensitivity to prednisolone or any of the excipients.

Peptic ulcer, osteoporosis, psychoses or severe psychoneuroses. Patients with active or doubtfully quiescent tuberculosis should not be given these hormones except as adjuncts to treatment with tuberculostatic drugs.

WARNINGS & PRECAUTIONS:

Pheochromocytoma crisis, which may be fatal, has been reported after administration of systemic corticosteroids. Corticosteroids should only be administered to patients with suspected or identified pheochromocytoma after as appropriate risk/ benefit evaluation.

Corticosteroids should be used with caution in the presence of diminished cardiac reserve or congestive heart failure, in patients with diabetes mellitus, infectious diseases, chronic renal failure, uraemia and in elderly persons.

In patients on corticosteroid therapy subjected to unusual stress, increased dosage of rapidly acting corticosteroids before, during, and after the stressful situation is indicated. Corticosteroids may mask some signs of infection, and new infections may appear during their use. Infections with any pathogen including viral, bacterial, fungal, protozoan or helminthic infections, in any location of the body, may be associated with the use of corticosteroids alone or in combination with other immunosuppressive agents that affect cellular immunity, humoral immunity, or neutrophil function. These infections may be mild, but can be severe and at times fatal. With increasing doses of corticosteroids, the rate of occurrence of infectious complications increases. There may be decreased resistance and inability to localize infection when corticosteroids are used.

Prolonged use of corticosteroids may produce posterior subcapsular cataracts, glaucoma with possible damage to the optic nerves, and may enhance the establishment of secondary ocular infections due to fungi or viruses.

Persons who are on drugs which suppress the immune system are more susceptible to infections than healthy individuals. Chicken pox and measles, for example, can have a more serious or even fatal course in non-immune children or adults on corticosteroids. In such children or adults who have not had these diseases, particular care should be taken to avoid exposure. How the dose, route and duration of corticosteroid administration affect the risk of developing a disseminated infection is not known. The contribution of the underlying disease and/or prior corticosteroid treatment to the risk is also not known.

Withdrawal symptoms: During prolonged treatment with corticosteroids, adrenal suppression and atrophy may occur and secretion of corticotrophin may be suppressed. Sudden withdrawal of the hormone treatment may then precipitate acute adrenal insufficiency with muscle weakness, hypotension, hypoglycaemia, headache, nausea, vomiting, restlessness and muscle and joint pain. Muscle weakness and stiff joints may persist for three to six months after treatment has been discontinued. In some instances, withdrawal symptoms may stimulate a clinical relapse of the disease for which the patient has been under treatment. Duration of treatment and dosage appear to be important factors in determining suppression of the pituitary adrenal axis and response to stress on cessation of steroid treatment. Individual liability to depression is also variable. Some patients may recover normal function rapidly. In others, the production of hydrocortisone in response to the stress of infections, surgical operations or accident may be insufficient, and death results. Withdrawal of corticosteroids should therefore always be gradual but if sudden withdrawal is necessary, corticotrophin (20 units) given daily by intravenous infusion over eight hours for three to five successive days is usually sufficient to prevent withdrawal symptoms.

There is an enhanced effect of corticosteroids on patients with hypothyroidism and in those with cirrhosis. Corticosteroids should be used cautiously in patients with ocular herpes simplex because of possible corneal perforation.

The lowest possible dose of corticosteroid should be used to control the condition under treatment, and when reduction in dosage is possible, the reduction should be gradual. Psychic derangements may appear when corticosteroids are used, ranging from euphoria, insomnia, mood swings, personality changes, and severe depression, to frank psychotic manifestations. Also, existing emotional instability or psychotic tendencies may be aggravated by corticosteroids.

Steroids should be used with caution in nonspecific ulcerative colitis, if there is a probability of impending perforation, abscess or other pyogenic infection; diverticulitis; fresh intestinal anastomoses; active or latent peptic ulcer; renal insufficiency; hypertension; osteoporosis; and myasthenia gravis.

Growth and development of infants and children on prolonged corticosteroid therapy should be carefully observed. Kaposi's sarcoma has been reported to occur in patients receiving corticosteroid therapy. Discontinuation of corticosteroids may result in clinical remission. Although controlled clinical trials have shown corticosteroids to be effective in speeding the resolution of acute exacerbations of multiple sclerosis, they do not show that corticosteroids affect the ultimate outcome or natural history of the disease. The studies do show that relatively high doses of corticosteroids are necessary to demonstrate a significant effect.

Since complications of treatment with glucocorticoids are dependent on the size of the dose and the duration of treatment, a risk/benefit decision must be made in each individual case as to dose and duration of treatment and as to whether daily or intermittent therapy should be used.

Scleroderma renal crisis

Caution is required in patients with systemic sclerosis because of an increased incidence of (possibly fatal) scleroderma renal crisis with hypertension and decreased urinary output observed with a daily dose of 15mg or more prednisolone.

INTERACTION WITH OTHER MEDICAMENTS:

The following drug interactions with corticosteroids have been selected on the basis of their potential clinical significance: antacids, antidiabetic agents (oral or insulin), digitalis glycosides, diuretics, drugs which induce hepatic microsomal enzymes, such as barbiturates, phenytoin and rifampicin; potassium supplements, ritodrine, sodium-containing medications or foods, somatrem or somatropin, vaccines, live viruses or other immunizations.

Drugs that induce hepatic enzymes such as phenobarbital, phenytoin and rifampin may increase the clearance of corticosteroids and may require increases in corticosteroid dose to achieve the desired response.

Drugs such as troleandomycin and ketoconazole may inhibit the metabolism of corticosteroids and thus decrease their clearance. Therefore, the dose of corticosteroid should be titrated to avoid steroid toxicity. Corticosteroids may increase the clearance of chronic high dose aspirin. This could lead to decreased salicylate serum levels or increase the risk of salicylate toxicity when corticosteroid is withdrawn. Aspirin should be used cautiously in conjunction with corticosteroids in patients suffering from hypoprothrombinemia.

The effect of corticosteroids on oral anticoagulants is variable. There are reports of enhanced as well as diminished effects of anticoagulants when given concurrently with corticosteroids. Therefore, coagulation indices should be monitored to maintain the desired anticoagulant effect.

Effects on Laboratory tests: Glucocorticoids may decrease ¹³¹I uptake and protein-bound iodine concentrations, making it difficult to monitor the therapeutic response of patients receiving the drugs for thyroiditis. Glucocorticoids may produce false-negative results in the nitroblue tetrazolium test for systemic bacterial infection. Glucocorticoids may suppress reactions to skin tests.

Convulsions have been reported with concurrent use of methylprednisolone and cyclosporin. Since concurrent use of these agents results in a mutual inhibition of metabolism, it is possible that adverse events associated with the individual use of either drug may be more apt to occur.

Co-treatment with CYP34 inhibitors, including cobicistat-containing products, is expected to increase the risk of systemic side-effects. The combination should be avoided unless the benefit outweighs the increased risk of systemic corticosteroid side effects, in which case patients should be monitored for systemic corticosteroid side-effects.

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: In animal experiments, corticosteroids have been found to cause malformations of various kinds (cleft palate, skeletal malformations) and abortion. These findings do not seem to be relevant to humans. Reduced placental and birth weight have been recorded in animals and humans after long term treatment. Since the possibility of suppression of the adrenal cortex in the newborn infant after long-term treatment must be considered, the needs of the mother must be carefully weighed against the risk to the fetus when prescribing these drugs. The short-term use of corticosteroids antepartum for the prevention of respiratory distress syndrome does not seem to pose a risk to the fetus or the newborn infant. Maternal pulmonary oedema has been reported with tocolysis and fluid overload.

Use in Lactation: The drug is excreted in breast milk; therefore, administration to nursing mothers is not recommended.

SIDE EFFECTS:

Fluid and Electrolyte disturbances: Sodium retention, fluid retention, congestive heart failure in susceptible patients, potassium loss, hypokalemic alkalosis, hypertension.

Musculoskeletal: Muscle weakness, steroid myopathy, loss of muscle mass, osteoporosis, tendon rupture, particularly of the Achilles tendon, vertebral compression fractures, aseptic necrosis of femoral and humeral heads, pathologic fracture of long bones.

Gastrointestinal: Peptic ulcer with possible perforation and hemorrhage, pancreatitis, abdominal distention, ulcerative oesophagitis

Increases in alanine transaminase (ALT), aspartate transaminase (AST) and alkaline phosphatase have been observed following corticosteroid treatment. These changes are usually small, not associated with any clinical syndrome and are reversible upon discontinuation.

Dermatologic: Impaired wound healing, thin fragile skin, Petechiae and ecchymoses, facial erythema, increased sweating, may suppress reactions to skin tests

Metabolic: Negative nitrogen balance due to protein catabolism

Neurological: Increased intracranial pressure with papilloedema (pseudo-tumour cerebri) usually after treatment, convulsions, vertigo, headache

Endocrine: Menstrual irregularities, development of Cushingoid state, secondary adrenocortical and pituitary unresponsiveness, particularly in times of stress, as in trauma, surgery or illness. Suppression of growth in children, decreased carbohydrate tolerance, manifestations of latent diabetes mellitus, increased requirements for insulin or oral hypoglycaemic agents in diabetics

Ophthalmic: Posterior subcapsular cataracts, increased intraocular pressure, glaucoma, exophthalmos

Additional Reactions: Urticaria and other allergic, anaphylactic or hypersensitivity reactions

SYMPTOMS AND TREATMENT OF OVERDOSE:

Treatment of toxic effects of corticosteroids: There is no specific antidote for the corticosteroids. Toxic effects are signs of overdosage, and should be treated symptomatically and dosage reduced or the drug withdrawn. During long courses of treatment, laboratory and metabolic studies should be made; fluid retention should be watched for via a fluid balance chart and daily weighing. Sodium intake may need to be reduced to less than 1 g daily and potassium supplements may be necessary.

STORAGE CONDITIONS:

Store below 30°C. Protect from light. Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

SHELF LIFE:

Please refer to outer package.

PACK SIZE:

Blister pack of 100 x 10's tablets/ box.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

Lot 2599, Jalan Seruling 59, Kawasan 3,

Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA